



Unit 3 · Outcome 1 · Maintaining Biodiversity

Reading the shape of life from the rocks

KK04

VCE Environmental Science · Units 3 & 4 · Exam study summary

Biodiversity hasn't crept upward at a steady pace. The fossil record shows life climbing, crashing, and exploding again — five catastrophic mass extinctions and several bursts of rapid diversification. This is how environmental scientists read the rate and extent of that change straight out of layered rock.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



HOW WE KNOW

The fossil record is a torn flip-book

Every claim in this dot point — that diversity rose, crashed, or exploded — comes from one source of evidence: the fossil record. Fossils form when dead organisms are buried in sediment that hardens into rock. Layer piles on layer, so the column of rock at Wirra Gorge is a stack of time.

◆ ANALOGY

Analogy · the torn flip-book Imagine a flip-book animation of life on Earth. The fossil record is that book — but with most pages torn out. Soft-bodied animals (jellyfish, worms) rarely survive as fossils. Erosion shreds whole chapters. So we never see every species; we see a minimum sample. When palaeontologists say "85% of species went extinct," they're reconstructing the missing pages from the few that survived. That's why the record is evidence to be inferred from, not a complete census.

Undisturbed layers stack oldest-at-the-bottom, youngest-on-top. Depth $\approx$ age. A fossil 40 m down is older than one near the surface.

Relative: this layer is older than that one. Absolute: radiometric dating of isotopes gives an actual age in years ($\pm$ an error).

A group's first appearance in the rock marks roughly when it arose; its last appearance marks extinction. Count taxa per layer $\rightarrow$ a diversity curve.

• NOTE

The boundary clay. At exactly the dinosaur-killing layer sits a thin, world-wide band of clay rich in iridium — an element rare in Earth's crust but common in asteroids. Finding the same chemical signature on every continent is powerful evidence that one global event, not gradual change, drew the line. This is "inferring from the record" in action.

THE TWO WORDS EXAMINERS REWARD

"Rate" and "extent" are not the same axis

The dot point asks about variation in rate and extent. Students who blur these together lose easy marks. They answer two different questions about any change in biodiversity.

How fast biodiversity is changing — species lost or gained per unit of time. A mass extinction is a spike in the rate of extinction. Steady, low-level loss is the background rate.

How much — the magnitude of the change, usually as a percentage of taxa affected. End-Permian's extent (~96% of marine species) dwarfs a minor extinction's extent.

◆ ANALOGY

Analogy · speed vs distance Rate is the speedometer; extent is the odometer. Two extinctions can share a rate (both fast) but differ wildly in extent (one wipes 20%, the other 96%). When a question says describe the variation in rate and extent, name both: how quickly and how severely.

The normal, continuous, low rate at which species go extinct as part of ongoing evolution and environmental change — roughly 1 species per million species per year. A mass extinction is when the rate climbs far above this baseline and the extent passes ~75% of species in a geologically short interval.

WHEN THE RATE SPIKES

The "Big Five" mass extinctions

Palaeontologists Raup & Sepkoski (1982) plotted marine families through time and found five moments where the extinction rate rose far above background. A mass extinction is defined as the loss of roughly $\geq 75\%$ of species in a geologically short interval (typically under ~ 2 million years).

• NOTE

The thread that links four of the five: rapid climate change. Warming (end-Permian and end-Triassic, from massive volcanism) or cooling (end-Ordovician and Late Devonian, from glaciation) destabilised ecosystems faster than species could adapt. Only the end-Cretaceous had a clear extra-terrestrial trigger — the Chicxulub asteroid — though Deccan Traps volcanism likely worsened it. This is the bridge to your Climate Change unit (U4.O1).

◆ ANALOGY

Analogy · the difficulty spike Think of evolution as a game balanced for the "background" difficulty — species adapt about as fast as conditions shift. A mass extinction is the difficulty slider suddenly slammed to maximum: the environment changes faster than evolution can keep up, so even well-adapted, abundant species are wiped. Survival becomes less about being "the best" and more about luck and tolerance — generalists with wide ranges scrape through; hyper-specialists don't.

WHEN THE RATE SPIKES THE OTHER WAY

Rapid diversification fills the empty world

Extinction is only half the dot point. The fossil record also shows the opposite spike: rapid species diversification (also called adaptive radiation), where the number of species climbs steeply in a short interval. Two situations trigger it.

A mass extinction vacates ecological niches. Surviving lineages, now free of competitors and predators, radiate to fill the space. The classic case: mammals exploded in diversity after the non-avian dinosaurs vanished at the end-Cretaceous.

A new trait or newly available environment opens opportunity. The Cambrian explosion (~ 541 Mya) saw most major animal body plans appear in a geological blink as hard shells, eyes and predation evolved.

This lab makes the cause-and-effect visible. Hit Trigger mass extinction and watch most branches of the tree of life die back. Then hit Let survivors radiate — the lineages that scraped through burst into new branches, filling the niches the dead left behind. That sequence is the single most exam-rewarded idea in this dot point.

• NOTE

Why this matters for the "rate" of change. Recoveries are not instant. After the end-Permian it took over 10 million years for diversity to rebuild. So the record shows a fast crash followed by a slower climb — a sawtooth, not a symmetric V. Examiners love a student who notes that extinction can be faster than recovery.

PUTTING THE CURVE TOGETHER

Life's diversity: a jagged climb, not a smooth ramp

Stand back from 540 million years and a pattern emerges. Overall biodiversity has broadly increased, but the line is punctuated — sharp crashes (the Big Five and many smaller events) and steep recoveries. The rate and extent of change are anything but constant. That variability is the dot point.

◆ ANALOGY

Analogy · the sharemarket of life A long-run stock chart trends upward but is jagged — booms (radiations) and crashes (extinctions). The Big Five are the great crashes; recoveries are the rallies. And the chart never returns to exactly the same companies: after each crash, new "stocks" dominate. After the end-Cretaceous crash, the dinosaur "blue chips" were delisted and the mammal "start-ups" took over the index — which is why a Helmeted Honeyeater, not a small theropod, sings at Kestrel Ridge today.

Every species you'll try to conserve in this chapter is a survivor of five mass extinctions — the current end-product of life's jagged climb. The "sixth extinction" framing matters because the threats in KK10 (habitat loss, climate change, introduced species) are pushing today's extinction rate toward mass-extinction territory. Deep time gives conservation its urgency: we may be the asteroid.

ANSWER THE VERB THAT'S ACTUALLY ASKED

Command-word decoder

◆ COMMAND WORDS

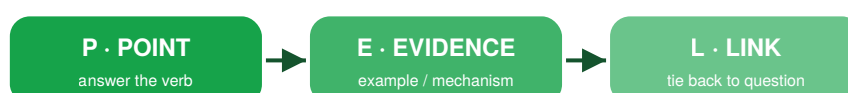
The same content earns different marks depending on the command word. Tap each to see what the examiner wants — and a worked sentence using this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

STRUCTURE THAT SCORES

P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P.E.L. WRITING

Extended-response marks come from structure as much as facts. Build your answer as Point → Evidence → Link. Type a response to the prompt below; the meters light up as your writing hits each element, and detected key terms appear underneath.

EXAM REHEARSAL · 17 MARKS

Practice questions — with weak & strong responses

Each question shows a weak and a strong response so you can see why marks are earned. Open the marking guide, mark your own attempt honestly, and the dock tracks your total.

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"Evidence of variation in rate and extent of change in biodiversity over time: significant mass extinctions and periods of rapid species diversification that can be inferred from the fossil record."

• NOTE

The running site for this chapter. Back in KK01 we mapped today's biodiversity at Kestrel Ridge Reserve, a box-ironbark woodland in central Victoria. This module zooms out to deep time. On the reserve's edge a disused quarry, Wirra Gorge, slices through 540 million years of stacked sediment. Your Year 12 class is going to read it like a book — bottom page first.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.